Wealth Inequality and Sudden Stops

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Wealth Inequality as a Determinant of Macroeconomic Vulnerability

- Abrupt halts in capital flows coincide with deep recessions in open economies
- Sudden stops with empirical regularities (Calvo, 1998)
  - Current account reversals
  - Contractions in output and domestic absorption
  - Drops in asset prices
- Asset buffers shape how households deleverage and cut spending during sudden stops
- Wealth distribution shapes how individual adjustments translate into aggregate contraction
- This paper: Does wealth inequality deepen recessions in sudden stops?

Preview of Findings

1. **Empirical findings**
   - Greater wealth inequality is associated with larger contractions during sudden stops (Median to 75th percentile: estimated consumption decline **0.31%** $\rightarrow$ **0.57%** per 1pp larger capital outflow/GDP)
   - More unequal countries are more leveraged
   - Indebted households cut debt and asset purchases sharply
2. **Tractable model** with an exogenous initial wealth distribution
   - Inequality increases mass of borrowing-constrained households
   - Asset price and consumption fall more after the same negative aggregate shock
3. **Quantitative model** with evolution of wealth distribution
   - 0.01 increase in wealth Gini adds 0.11pp to consumption drops per 1pp rise in capital outflow/GDP
4. **Normative analysis**
   - Joint taxes on wealth and debt insulate households from becoming constrained

Literature Review

- Sudden stops: Calvo, 1998, Mendoza, 2010, Bianchi, 2011, Castillo-Martinez, 2025, Miranda-Pinto et al., 2025, Villalvazo, 2026

  This paper: inequality amplifies recessions and contractions of asset demand in sudden stops

- Macroprudential regulation: Bianchi, 2011, Bianchi and Mendoza, 2018, Korinek, 2018, Biljanovska and Vardoulakis, 2024

  This paper: jointly optimal wealth and debt taxes with a continuous wealth distribution

- Inequality and aggregate dynamics: Galor and Zeira, 1993, Persson and Tabellini, 1994, Mian et al., 2021, Krusell and Smith, 1998, Kaplan et al., 2018, Auclert, 2019, Bayer et al., 2024, Kumhof et al., 2015, Paul, 2023

  This paper: wealth inequality shapes the severity of financial crises

- Household heterogeneity in open economies: de Ferra et al., 2020, Auclert et al., 2021, Oskolkov, 2023, Ferrante and Gornemann, 2025, Guntin et al., 2023, Verner and Gyöngyösi, 2020

  This paper: household-level adjustments in wealth around sudden stops

Data

- Annual panel of 54 countries, 1979–2016
  - Macroeconomic data from IMF IFS and Household net wealth Gini from World Inequality Database
- Sudden stop definition (Korinek and Mendoza, 2014, Bianchi and Mendoza, 2020)
  - Change in CA/GDP exceeds its country mean by 2 standard deviations
  - VIX (advanced) or EMBI spreads (emerging) exceed their mean by 1 standard deviation
- ECB HFCS: repeated cross-section of households from 15 countries in 2011 and 2014
  - Greece, Italy, Portugal, and Spain experienced sudden stops in 2012
  - Compare changes in wealth of household groups before and after these episodes

▸ Episodes

Fact 1: Larger Contractions under Greater Wealth Inequality

- Sudden-stop intensity: Net Flow$^{SS} \equiv SS \times$ Net Capital Outflows/GDP
- Continuous inequality index $\nu_{i,t-1}$: 0 at the median wealth Gini, 1 at the 75th percentile

$$Y_{it} = \alpha + \beta \text{Net Flow}_{it}^{SS} + \gamma \text{Net Flow}_{it}^{SS} \times \nu_{i,t-1} + \Gamma X_{i,t-1} + FE + \epsilon_{it}$$

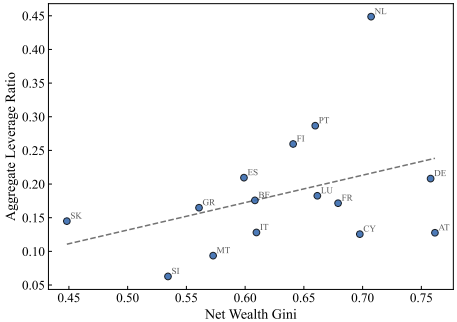
	GDP	Consumption	Investment	Stock prices
Net Flow^{SS} (β)	-0.154^{***}	-0.314^{***}	-1.084^{***}	-1.877^{***}
	(0.047)	(0.031)	(0.170)	(0.511)
Net Flow$^{SS} \times \nu$ (γ)	-0.309^{***}	-0.252^{***}	-1.414^{***}	-1.821^{**}
	(0.064)	(0.045)	(0.175)	(0.816)
Controls	Yes	Yes	Yes	Yes
Observations	864	864	847	782
Within R^2	0.531	0.432	0.357	0.085

▸ Controls ▸ Inequality Measure ▸ Dynamics ▸ T10/B50 ▸ Income Inequality ▸ Other Interactions ▸ Subsamples ▸ Emerging Markets ▸ Normal Times ▸ Other Crises

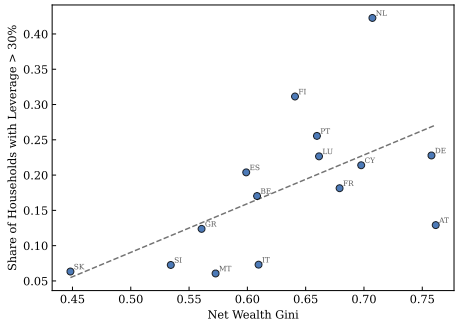
Fact 2: More Unequal Economies Are More Leveraged

- Higher leverage on aggregate and individual levels in more unequal countries
- External debt is larger in more unequal countries

▸ External Debt Regression



(a) Aggregate household debt / assets



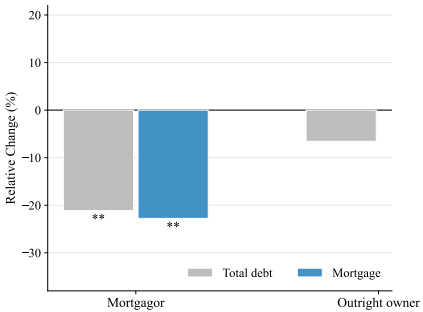
(b) Fraction of high-leverage households

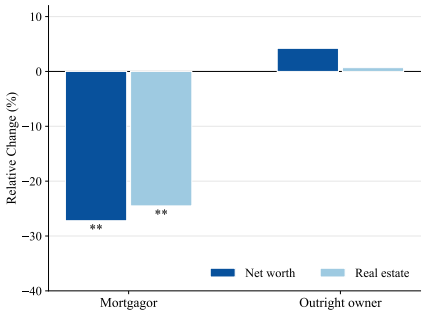
Fact 3: Indebted Households Deleverage

- Repeated cross-sections: mortgage status proxies for indebtedness

- $\Delta y_{c,g}$: change in asset category y's mean for group $g \in \{\text{Indebted}, \text{Not Indebted}\}$ in country c

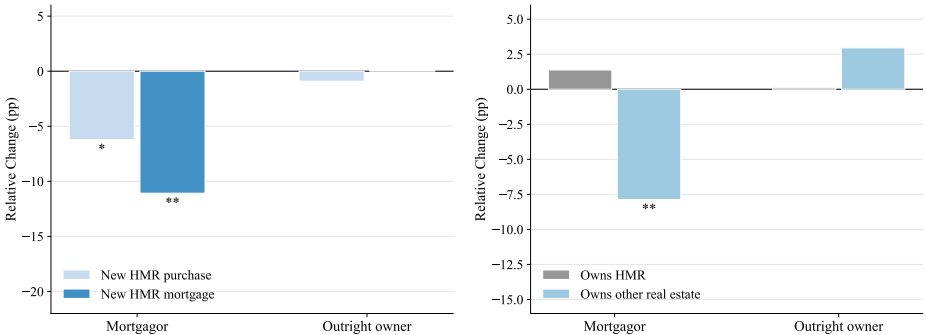
$$\delta_g = \frac{1}{4} \sum_{c \in \text{SS}} \Delta y_{c,g} - \frac{1}{10} \sum_{c \notin \text{SS}} \Delta y_{c,g}, \quad \text{SS} = \{\text{GR, IT, PT, ES}\}$$





Fact 3: Indebted Households Reduce Asset Purchases

- New mortgage origination and new home purchases decrease for indebted households
- Indebted households also hold less real estate other than main residences



Takeaway: Empirical Analysis

- Greater wealth inequality is associated with larger contractions and asset price drops
- More unequal countries enter crises more leveraged
- Balance-sheet reductions and lower asset purchases concentrated among indebted households

  $\Rightarrow$ Model: how inequality raises the mass of constrained households

Model Environment

- Three-period model with initial wealth distribution: $k_{i0} \sim G$
  - Greater wealth inequality modeled as a mean-preserving spread of initial distribution
- No idiosyncratic shock to households to shut down the role of income risk
- Households subject to loan-to-value borrowing constraint with constant θ
- Two types of assets
  - domestic capital at price q_t that clears domestic capital market: $\int k_{i,t+1} di = \int k_{i0} di$
  - foreign bonds at exogenous interest rate R
- Focus on effect of negative TFP shock in second period and its amplification by inequality

Individual Household's Problem

- Household i begins with initial assets k_{i0} and maximizes lifetime utility defined as

$$\max_{\substack{\{c_{it},k_{i,t+1},b_{i,t+1}\}_{t=0,1} \\ c_{i2}}} \mathbb{E}_0 \left[\log c_{i0} + \beta \log c_{i1} + \beta^2 c_{i2} \right]$$

$$\begin{aligned} \text{s.t. } & c_{i0} + q_0 k_{i1} + \frac{b_{i1}}{R} = f(k_{i0}) + q_0 k_{i0} \\ & c_{i1} + q_1 k_{i2} + \frac{b_{i2}}{R} = z_1 f(k_{i1}) + q_1 k_{i1} + b_{i1} \\ & c_{i2} = f(k_{i2}) + k_{i2} + b_{i2} \\ & \frac{b_{i,t+1}}{R} \geq -\theta q_t k_{i,t+1}, \quad t \in \{0, 1\} \end{aligned}$$

  where $u(c) = \log c$, $f(k) = k^\alpha$, $\alpha \in (0, 1)$, and $\beta R = 1$

- z_1 is stochastic from period 0's perspective, and $z_0 = z_2 = 1$

Market Clearing and Resource Constraint

- Aggregate consumption, foreign bonds, and output:

$$C_t \equiv \int c_{it}\, di, \qquad B_t \equiv \int b_{it}\, di, \qquad Y_t \equiv \int z_t f(k_{it})\, di$$

- Fixed aggregate capital; q_t clears the domestic capital market:

$$\int k_{i,t+1}\, di = \int k_{it}\, di = K, \qquad t = 0, 1$$

- Aggregating household budgets gives the resource constraint:

$$C_t + \frac{B_{t+1}}{R} = Y_t + B_t, \qquad t = 0, 1, \quad B_0 = 0$$

- In the terminal period, capital is consumed after production:

$$C_2 = Y_2 + K + B_2$$

TFP Shock Analysis in Period 1

- Period-1 wealth: $\tilde{w}_i = z_1 f(k_{i1}) + q_1 k_{i1} + b_{i1}$
- Unconstrained households ($\mathcal{U}$) demand $\overline{k}_2 = (f')^{-1}(q_1 R - 1)$
- Endogenous threshold: $\overline{w} = 1 + (1 - \theta) q_1 \overline{k}_2$
- Constrained households ($\mathcal{C}$) have $\tilde{w}_i < \overline{w}$ and demand $k_{i2} < \overline{k}_2$

Asset Price Response to a TFP Shock

- Lower z_1 reduces constrained households' capital demand at a given price
- Response of asset price to TFP shock depends on two components:

$$\frac{dq_1}{dz_1} = \frac{\overbrace{\int_{\mathcal{C}} \frac{\partial k_{i2}}{\partial z_1} di}^{\text{Direct contraction in constrained HH's demand}}}{\underbrace{\int_{\mathcal{C}} \left| \frac{\partial k_{i2}}{\partial q_1} \right| di + \int_{\mathcal{U}} \left| \frac{\partial \overline{k}_2}{\partial q_1} \right| di}_{\text{Aggregate price elasticity of capital demand}}} > 0$$

- Contraction in asset demand comes from constrained HHs (Fact 3)

Effects of Negative TFP Shock

- Income losses reduce wealth, while lower prices raise $\overline{w}$

▸ Lemma 1

- More households become constrained and cut consumption

Proposition

A negative shock to z_1

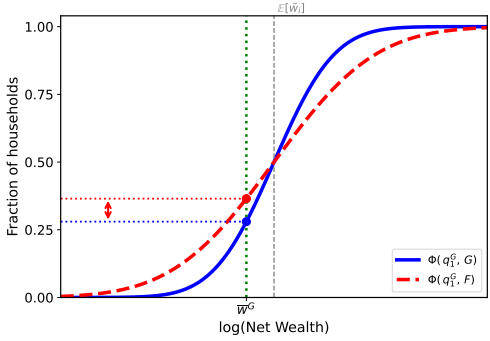
1. *lowers the asset price q_1*
2. *lowers aggregate consumption C_1*

$$\frac{dC_1}{dz_1} = \underbrace{\int_{\mathcal{C}} \frac{\partial c_{i1}}{\partial z_1}\bigg|_{q_1} di}_{\text{PE Effects}>0} + \underbrace{\int_{\mathcal{C}} \frac{\partial c_{i1}}{\partial q_1}\frac{dq_1}{dz_1} di}_{\text{GE Effects}>0}$$

3. *generates a capital outflow if many HHs are constrained with low TFP: $z_1 < z^*$*

Wealth Inequality: Constrained Share at a Fixed Price

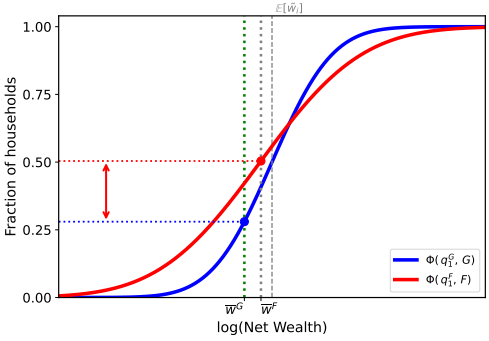
- Partial equilibrium: hold the asset price fixed at q_1^G
- A mean-preserving spread F places more households in the lower tail than G
- More households lie below the same borrowing threshold $\overline{w}^G$



More households at the borrowing limit, consistent with higher leverage (Fact 2)

Wealth Inequality: Equilibrium Price Adjustment

- General equilibrium: the asset price adjusts
- At q_1^G, lower demand under F requires a lower equilibrium price q_1^F
- Lower wealth and $\overline{w}^F > \overline{w}^G$ raise the constrained share



▸ Lemma 1

Wealth Inequality Amplifies the Contraction

- Mean-preserving spread: $\Phi(x; q_1, \Phi_\lambda) = \lambda\Phi(x; q_1, G)$ for $x < \mathbb{E}_G[\tilde{w}]$
- $\lambda \uparrow$: greater inequality and more constrained HHs

Proposition

If $R\theta q_1^G \leq 1$ *and* $\overline{w}^G < \min\{\mathbb{E}_G[\tilde{w}], Med_G[\tilde{w}]\}$, *then*

1. *The asset price falls more for the same* z_1 *shock:*

$$\frac{\partial^2 q_1}{\partial\lambda\,\partial z_1} > 0 \qquad \frac{\textit{Average direct effect of shock} \uparrow}{\textit{Average elasticity of asset demand} \downarrow}$$

▸ Lemma 2

2. *If* $\alpha \geq 1/2$, *consumption also falls more:*

$$\frac{\partial^2 C_1}{\partial\lambda\,\partial z_1} > 0$$

Numerical Results under a Productivity Shock

- Effects of negative TFP shock under two initial wealth distributions with same mean
- Drops in consumption and asset price are larger under greater initial inequality

	Baseline Distribution (G)	Alternative Distribution (F)
Wealth Gini at period 0	0.61	0.70
Mean wealth at period 1 when $z_1 = z_H$	5.08	5.02
Wealth Gini at period 1 when $z_1 = z_H$	0.57	0.64
Constrained fraction when $z_1 = z_H$ (%)	37.69	49.54
Constrained fraction when $z_1 = z_L$ (%)	39.73	51.58
Change in aggregate consumption C_1 (%)	-1.62	-2.13
Change in asset price q_1 (%)	-0.31	-0.40

▸ Decomposition ▸ Decomposition Results

Takeaway: Tractable Model

- Greater inequality raises the fraction of constrained households
- Constrained households cut asset purchases after the shock
- Less elastic demand of assets amplifies price and consumption drops
- Numerical examples: amplification also occurs for θ and R shocks

  $\Rightarrow$ Quantitative model: endogenous evolution of wealth inequality

▸ Other Shocks

Quantitative Model

- I extend the model to an infinite horizon economy with
  - endogenous evolution of the joint distribution Γ of (k, b, ω)
  - heterogeneous entrepreneur-households with permanently different productivity ω
  - capital accumulation through representative capital producers
- Individual household's problem with aggregate state $s = (z, R, \theta, \Gamma)$:

$$V(s, k, b; \omega) = \max_{c,n,k',b'} u(c - \nu(n)) + \beta \mathbb{E} V(s', k', b'; \omega)$$

$$\text{s.t. } c + qk' + \frac{\zeta}{2}(k' - k)^2 + \frac{b'}{R} = z\omega f(k, n) + q(1 - \delta)k + b + \pi$$

$$\frac{b'}{R} \geq -\theta q k'$$

Quantitative Model: Capital Producers

- Representative capital producers subject to quadratic production costs:

$$\max_i \pi = qi - i - \frac{\phi}{2}\frac{i^2}{K}$$

which implies that the capital price satisfies $q = 1 + \phi \frac{I}{K}$ in equilibrium

Parameter Calibration

- GHH utility function:

$$u(c - \nu(n)) = \frac{\left(c - \frac{n^{1+\frac{1}{\psi}}}{1+\frac{1}{\psi}}\right)^{1-\sigma}}{1-\sigma}$$

- TFP z and interest rate R follow AR(1) processes

- $\theta \in \{\theta_H, \theta_L\}$ follows a two-state stationary Markov chain

$$\Pi_{\theta'|\theta} = \begin{pmatrix} p_{HH} & 1 - p_{HH} \\ 0.999 & 0.001 \end{pmatrix}$$

- Individual productivity ω is log-normal, normalized so that $\mathbb{E}(\omega) = 1$

- Estimate $\{\zeta, \theta_H, \theta_L, p_{HH}, \sigma_\omega\}$ to match 6 moments from data

▸ Calibration

Calibration Targets and Model Fit

- Krusell and Smith, 1998 method: 5,000 households, 20,000 periods

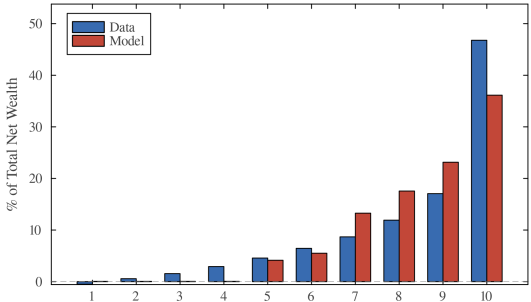
$$\log K' = 0.041 + 0.939 \log K + 0.114 \log z - 0.092 \log R - 0.052 \, \mathbb{I}(\theta = \theta_L), \quad R^2 = 0.979$$

Moment	Model	Data	Source
Average debt/GDP	27.62%	27.67%	Lane and Milesi-Ferretti, 2018
Sudden stop occurrence rate	3.65%	3.57%	Regression sample
Average wealth Gini	0.67	0.77	World Inequality Database
$\sigma_C/\sigma_{\text{GDP}}$	1.073	1.13	IMF IFS
$\sigma_I/\sigma_{\text{GDP}}$	1.589	3.95	IMF IFS
$\sigma_{\text{CA/GDP}}$	5.22%	3.85%	IMF IFS

Wealth Distribution in the Model and Data

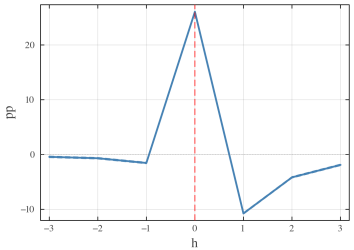
- Untargeted net wealth distribution from the model is close to the data

▸ Leverage Ratio

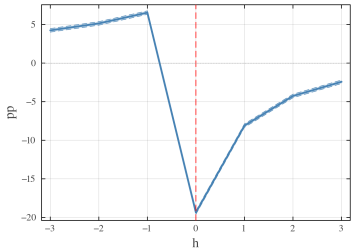


Sudden Stop Dynamics: Capital Flows

- Sudden stops: change in CA/GDP is larger than mean by 2 standard deviations
- Aggregate shocks: low z, rise in R, drop in θ around sudden stops



(a) Current Account to GDP

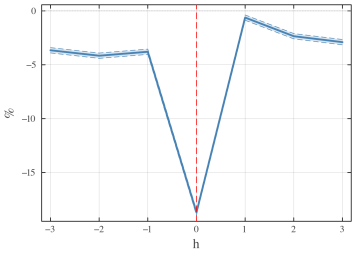


(b) Debt to GDP

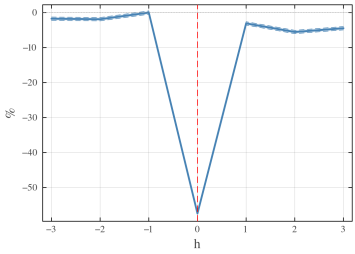
▸ Shocks ▸ Distributional Effects

Sudden Stop Dynamics: Spending and Asset Prices

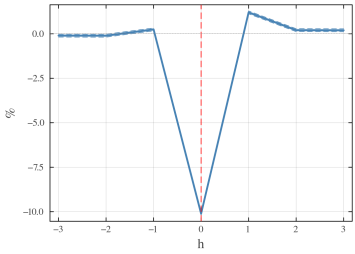
- Lower θ and falling asset prices tighten borrowing limits and reduce spending



(a) Consumption



(b) Investment

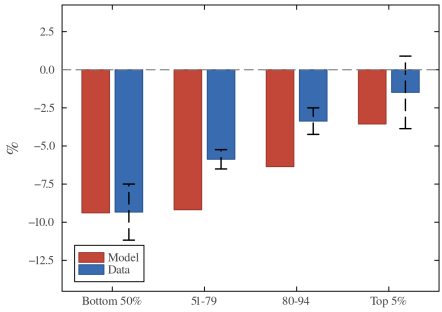


(c) Asset Price

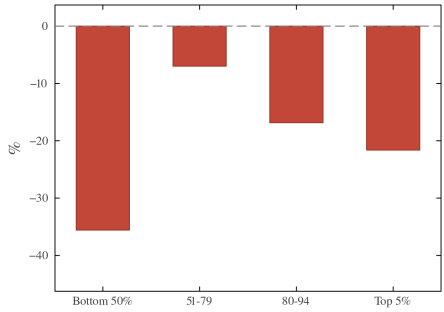
▸ Shocks ▸ Distributional Effects

Distributional Effects of Sudden Stops

- Net wealth falls more for poorer households in both the data and the model
- Model consumption cuts reflect small buffers at the bottom and debt repayment at the top



(a) Net wealth changes in the data and model



(b) Consumption changes in the model

Inequality and Recessions in Sudden Stops

- Nine economies: vary σ_ω, holding mean productivity, other parameters, and shocks fixed
- Inequality$_{t-1}$: lagged wealth Gini minus own median, in 0.01 units
- All three model estimates are within 95% confidence intervals of data counterparts

	Model			Data		
	GDP	C	I	GDP	C	I
Net Flow$_t^{SS}$	-0.106***	-0.638***	-2.308***	-0.084	-0.256***	-0.796**
	(0.001)	(0.010)	(0.010)	(0.095)	(0.064)	(0.354)
Net Flow$_t^{SS}$ $\times$ Inequality$_{t-1}$	-0.074***	-0.105***	-0.449***	-0.086*	-0.070**	-0.365**
	(0.006)	(0.010)	(0.037)	(0.051)	(0.026)	(0.155)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
FE	Economy	Economy	Economy	Cty, Year	Cty, Year	Cty, Year
Within R^2	0.492	0.583	0.408	0.518	0.426	0.339

Extensive Margin Effects of Inequality

- Greater productivity dispersion raises borrowing for consumption at the bottom and investment at the top
- Across these economies, more debt precedes more frequent sudden stops
- Wealth Gini 0.586 $\rightarrow$ 0.659: sudden stop probability 3.54% $\rightarrow$ 3.91%

	Extensive Margin		Intensive Margin				
Wealth Gini	Pr(SS) (%)	Debt/GDP (%)	ΔCA/GDP (pp)	ΔGDP (%)	ΔC (%)	ΔI (%)	Δq (%)
0.586	3.54	39.28	26.73	-4.74	-16.95	-54.16	-9.70
0.621	3.79	40.27	27.42	-4.42	-18.07	-55.45	-9.82
0.652	3.86	41.51	28.25	-4.39	-19.50	-57.96	-10.07
0.659	3.91	41.83	28.48	-4.30	-19.79	-58.58	-10.12

Government Problem

- Choose wealth and debt taxes under commitment, with revenues rebated uniformly
- Joint taxation (τ_t^w, τ_t^b) vs. capital controls $(\tau_t^w = 0)$

▸ Tax FOCs

$$\max_{\substack{\{\tau_t^w, \tau_t^b, T_t, q_t\}_{t\geq 0} \\ \{c_{j,t}, n_{j,t}, k_{j,t+1}, b_{j,t+1}\}_{j,t\geq 0}}} \quad \mathbb{E}_0 \sum_{t\geq 0} \beta^t \int u(c_{j,t} - \nu(n_{j,t}))\, dj$$

$$\text{s.t.} \quad c_{j,t} + q_t k_{j,t+1} + \frac{\zeta}{2}(k_{j,t+1} - k_{j,t})^2 + (1 - \tau_t^b)\frac{b_{j,t+1}}{R_t}$$

$$= z_t \omega_j f(k_{j,t}, n_{j,t}) + q_t(1 - \tau_t^w)(1 - \delta)k_{j,t} + b_{j,t} + \Pi(\{k_{j,t}, k_{j,t+1}\}) + T_t,$$

$$\frac{b_{j,t+1}}{R_t} \geq -\theta_t q_t k_{j,t+1},$$

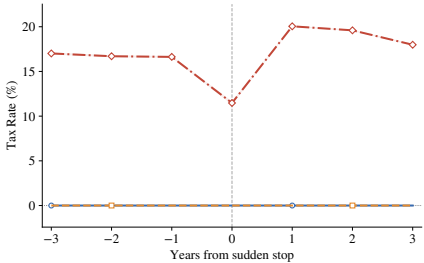
$$T_t = \tau_t^w q_t (1 - \delta) \int k_{j,t}\, dj - \tau_t^b \int \frac{b_{j,t+1}}{R_t}\, dj$$

Individual FOCs for $\forall j, t$ and capital market clearing condition

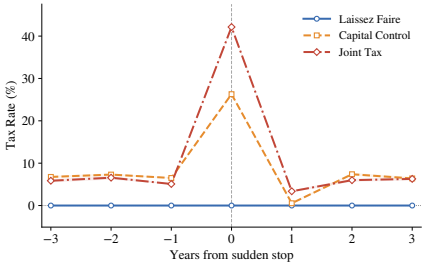
Optimal Taxes with Heterogeneous Households

- Joint taxation lowers the wealth tax in crises to preserve the return on capital carried into them
- Government relies more on debt taxation as wealth-tax revenue falls
- Capital controls: the crisis debt tax rises in HA and falls in RA

▸ RA–HA



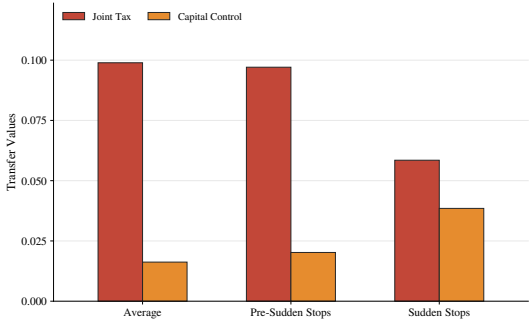
(a) Wealth tax



(b) External debt tax

Transfers and Borrowing Needs during Sudden Stops

- Under capital controls, transfers rise during sudden stops and support investment demand
- Under joint taxation, transfers fall but remain higher than under capital controls
- Higher transfer levels under joint taxation reduce households' need to borrow



Sudden Stop Dynamics under Taxation Policies

Deviation from ergodic mean	Laissez faire	Capital control	Joint taxation
Current account reversal (pp)	35.8	17.9	22.5
Consumption (%)	−24.2	−13.5	−18.2
Capital price (%)	−28.7	−14.1	−32.3
Constrained fraction (pp)	61.4	57.9	2.4

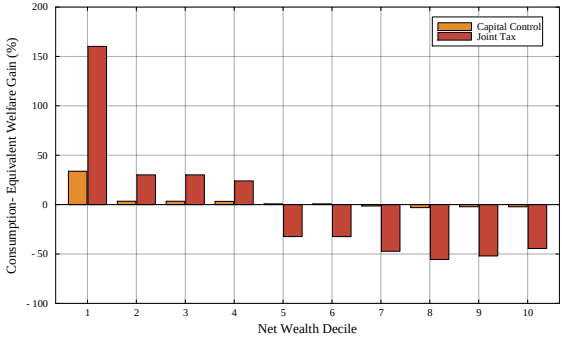
- Both policies lower pre-crisis debt and deleveraging pressure

▸ Figures

- Capital controls stabilize consumption and capital prices more
- Joint taxation limits the rise in constrained households despite the larger price decline
  - Larger transfers reduce households' need to borrow against collateral

Welfare Gains across the Wealth Distribution

- Consumption-equivalent gains: **11.24%** under joint taxes, **5.75%** under capital controls
- Taxation improves welfare of the bottom deciles at the expense of those at the top



Conclusion

- Greater wealth inequality is associated with larger contractions in sudden stops
  - Median to 75th percentile: estimated consumption decline **0.31%** $\rightarrow$ **0.57%** per 1pp larger capital outflow/GDP
- Households are more leveraged in more unequal countries
- Balance sheet adjustments are concentrated on indebted households
- Inequality endogenously increases fraction of constrained households
- Optimal taxes reduce rise in mass of constrained HHs in sudden stops: **61.4pp** $\rightarrow$ **2.4pp**

Thank you!

Income Inequality and Sudden Stops

- Baseline regression with income Gini coefficients as the inequality measure
- The interaction with income inequality is statistically insignificant for all four outcomes

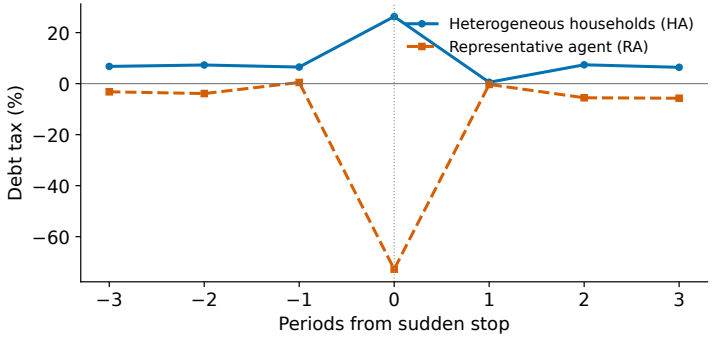
	GDP	Consumption	Investment	Δ Stock Price
Net Flow$_t^{SS}$	-0.205**	-0.276**	-1.323***	-2.351***
	(0.093)	(0.112)	(0.364)	(0.725)
Net Flow$_t^{SS} \times \nu_{t-1}^{\text{Income}}$	-0.034	0.068	-0.186	-0.323
	(0.045)	(0.089)	(0.115)	(0.308)
Observations	1229	1229	1207	981
Within R^2	0.501	0.400	0.298	0.064

- This motivates examining households' asset buffers and debt positions

Back

Capital Controls in RA and HA

- The debt tax falls into a subsidy in RA, but rises in HA during sudden stops



- HA adds redistribution from larger borrowers to households with a higher marginal value of wealth

Back

Price Decomposition: Constrained Share

$\mathcal{D}x = x^F - x^G$. Eq. (34), with its common positive multiplier omitted.

$$\mathcal{D}\left(\frac{dq_1}{dz_1}\right) \propto \underbrace{\frac{(1-\theta)q_{1,H}^G(\overline{k}_{2,L}^G - \overline{k}_{2,H}^G)}{(z_H - z_L)|\mathcal{C}_L^F||\mathcal{C}_L^G|}\mathcal{D}|\mathcal{C}_L|}_{\text{Constrained share}}$$

$$+ \underbrace{(1 - \mathbb{E}_{\mathcal{C}_L}^G[m_i^G])\left[\mathcal{D}\overline{F} - \frac{(dq_1/dz_1)^G}{q_{1,H}^G}\mathcal{D}\mathbb{E}[\tilde{w}_{i,H}]\right]}_{\text{Mean wealth / output}} + \underbrace{\mathcal{R}_q}_{\text{Household responses}} .$$

- **Constrained share ↑ ⇒ larger price decline**
  More households cut asset demand, and fewer unconstrained buyers absorb sales

◂ Numerical Results

Price Decomposition: Mean Wealth and Output

$\mathcal{D}x = x^F - x^G$. Eq. (34), with its common positive multiplier omitted.

$$\mathcal{D}\left(\frac{dq_1}{dz_1}\right) \propto \underbrace{\frac{(1-\theta)q_{1,H}^G(\overline{k}_{2,L}^G - \overline{k}_{2,H}^G)}{(z_H - z_L)|\mathcal{C}_L^F||\mathcal{C}_L^G|}\mathcal{D}|\mathcal{C}_L|}_{\text{Constrained share}}$$

$$+ \underbrace{(1 - \mathbb{E}_{\mathcal{C}_L}^G[m_i^G])\left[\mathcal{D}\overline{F} - \frac{(dq_1/dz_1)^G}{q_{1,H}^G}\mathcal{D}\mathbb{E}[\tilde{w}_{i,H}]\right]}_{\text{Mean wealth / output}} + \underbrace{\mathcal{R}_q}_{\text{Household responses}} .$$

- **Mean wealth and output have opposing effects**
  Lower output reduces income losses, while lower wealth weakens asset purchases

◂ Numerical Results

Price Decomposition: Household Responses

$\mathcal{D}x = x^F - x^G$. Eq. (34), with its common positive multiplier omitted.

$$\mathcal{D}\left(\frac{dq_1}{dz_1}\right) \propto \underbrace{\frac{(1-\theta)q_{1,H}^G(\overline{k}_{2,L}^G - \overline{k}_{2,H}^G)}{(z_H - z_L)|\mathcal{C}_L^F||\mathcal{C}_L^G|}\mathcal{D}|\mathcal{C}_L|}_{\text{Constrained share}}$$

$$+ \underbrace{(1 - \mathbb{E}_{\mathcal{C}_L}^G[m_i^G])\left[\mathcal{D}\overline{F} - \frac{(dq_1/dz_1)^G}{q_{1,H}^G}\mathcal{D}\mathbb{E}[\tilde{w}_{i,H}]\right]}_{\text{Mean wealth / output}} + \underbrace{\mathcal{R}_q}_{\text{Household responses}} .$$

- $\mathcal{R}_q$: **Distribution of MPC matters**
  High-MPC households absorb more of income loss through consumption cuts, so they reduce asset purchases less, mitigating the price decline

  High-MPC households devote less of additional purchasing power to assets, so lower prices induce less additional buying, amplifying the price decline.

◂ Numerical Results

The Constrained Share Drives Amplification

- Contribution to the larger decline in the more unequal economy

Additional decline (pp)	Asset price	Consumption
Constrained share	+0.15	+0.72
Average wealth and output	−0.01	−0.06
Household responses and prices	−0.05	−0.16
Total	+0.09	+0.50

- The larger share already constrained before the shock drives amplification
- Differences in resources and household responses partly offset it in this example

◂ Numerical Results

List of Variables

- Cyclical component of GDP per capita (lag)
- Wealth inequality (lag): continuous index ν
- Real exchange rate (lag)
- Trade openness (lag)
- Financial development index (lag)
- Debt to GDP from Lane and Milesi-Ferretti, 2018 (lag)
- International reserves (excluding gold) to GDP (lag)
- Chinn-Ito index of financial openness (lag)

Back

Construction of Wealth Inequality Measure

Following Auerbach and Gorodnichenko, 2017 and Lakdawala et al., 2021:

1. Standardize wealth Gini across the country-year sample

$$z_{it}^{Gini} = \frac{Gini_{it} - \overline{Gini}}{\sigma_{Gini}}$$

2. Apply the logistic transformation

$$\ell_{it} = \frac{1}{1 + \exp(-z_{it}^{Gini})}$$

3. Normalize relative to the median and the 75th percentile

$$\nu_{it} = \frac{\ell_{it} - \ell_{50\%}}{\ell_{75\%} - \ell_{50\%}}$$

Back

Comparison of Normal Times and Sudden Stops

- Separate Sudden Stops from ordinary times within the same specification

$$Y_{it} = \alpha + \sum_{k \in \{SS,\ \text{Non-SS}\}} \left(\beta^k \text{Net Flow}_{it}^k + \gamma^k \text{Net Flow}_{it}^k \times \nu_{i,t-1} \right) + \Gamma X_{i,t-1} + FE + \epsilon_{it}$$

	GDP	Consumption	Investment	Δ Stock Price
$\text{Net Flow}_t^{\text{Non-SS}}$	-0.038***	-0.040*	-0.355***	1.062**
	(0.013)	(0.024)	(0.071)	(0.480)
$\text{Net Flow}_t^{\text{Non-SS}} \times \nu_{t-1}$	-0.095***	-0.073***	-0.377***	-0.399
	(0.017)	(0.023)	(0.098)	(0.329)
Net Flow_t^{SS}	-0.148***	-0.308***	-1.044***	-1.927***
	(0.051)	(0.037)	(0.177)	(0.443)
$\text{Net Flow}_t^{SS} \times \nu_{t-1}$	-0.319***	-0.259***	-1.455***	-1.810**
	(0.052)	(0.035)	(0.132)	(0.819)
Observations	864	864	847	782
Within R^2	0.560	0.447	0.407	0.117

Back

Comparison to Other Crisis Episodes

- Bank runs (Jamilov et al., 2025) and systemic banking crises (Laeven and Valencia, 2020) were not selected on capital flows
- Other crises have average capital inflows of 3.34% of GDP. The GDP decline and its inequality interaction are significant only for sudden stops

	Sudden Stop		Bank Run (JKMS)		Banking Crisis (LV)	
	GDP	Δq	GDP	Δq	GDP	Δq
Crisis_t	-0.013**	-0.081	-0.010	-0.489*	-0.005	-0.590***
	(0.006)	(0.056)	(0.007)	(0.254)	(0.006)	(0.117)
$\text{Crisis}_t \times \nu_{t-1}$	-0.011*	-0.133*	-0.001	0.052	-0.002	0.065
	(0.006)	(0.078)	(0.006)	(0.126)	(0.006)	(0.148)
Observations	1128	941	1128	941	1128	941
Within R^2	0.527	0.064	0.518	0.112	0.515	0.154

Back

Using another measure of wealth inequality

- Baseline regression with top 10% to bottom 50% wealth ratio

	GDP	Consumption	Investment	Δ Stock Price
Net Flow$_t^{SS}$	-0.083	-0.251***	-0.820***	-1.105*
	(0.094)	(0.056)	(0.301)	(0.580)
Net Flow$_t^{SS}$ × T10/B50$_{t-1}$	-0.001***	-0.001***	-0.004***	-0.008**
	(0.000)	(0.000)	(0.001)	(0.004)
Observations	853	853	836	768
Within R^2	0.548	0.451	0.375	0.096

Back

Subsample Analysis

- High: Countries of which the average wealth Gini is above the cross-country median
  Low: Countries of which the average wealth Gini is below the cross-country median

	GDP		Consumption		Investment		Stock Price	
	High	Low	High	Low	High	Low	High	Low
Net Flow$_t^{SS}$	-0.710***	-0.057***	-0.739***	-0.232***	-3.739***	-0.616**	-4.401***	-1.381**
	(0.166)	(0.016)	(0.125)	(0.048)	(0.438)	(0.253)	(1.614)	(0.525)
Observations	391	471	391	471	374	471	375	407
Within R^2	0.555	0.467	0.456	0.387	0.415	0.287	0.060	0.137

Back

Sudden Stops in Emerging Markets

- Emerging markets and advanced economies differ in terms of financial development
- Study separately for emerging markets except for South Korea, which switched from EM to AE

	GDP	C	I	Δq
Net Flow$_t^{SS}$	-0.514**	-0.524***	-2.677***	-2.229
	(0.109)	(0.109)	(0.665)	(1.757)
Net Flow$_t^{SS} \times \nu_{t-1}$	-0.148	-0.182**	-0.668*	-2.449
	(0.132)	(0.083)	(0.333)	(1.456)
Observations	366	366	349	332
Within R^2	0.537	0.441	0.377	0.061

Back

Lemma 1: Borrowing Threshold

- Unconstrained demand $\overline{k}_2 = (f')^{-1}(q_1 R - 1)$ requires down payment $(1-\theta)q_1\overline{k}_2$

Lemma (Partial Effects of z_1 and q_1)

HHs with $\tilde{w}_i < \overline{w}$ are constrained where $\overline{w} = 1 + (1-\theta)q_1\overline{k}_2$, and

$$\underbrace{\frac{\partial \overline{w}}{\partial z_1} = 0}_{\text{no direct effect}} \quad \text{and} \quad \underbrace{\frac{\partial \overline{w}}{\partial q_1} < 0}_{\text{price drop raises threshold}} .$$

- z_1 changes households' wealth, but not the wealth needed to buy $\overline{k}_2$
- A lower price raises $\overline{k}_2$ more than proportionally, so the required down payment rises

◂ Proposition 1 ◂ Distribution

Lemma 2: Price Elasticity of Asset Demand

- Constrained households buy $k_{i2} < \overline{k}_2$ such that $(1-\theta)q_1 u'(c_{i1}) = \beta(f'(k_{i2}) + 1 - \theta q_1 R)$

Lemma

Constrained households' asset demand is less elastic than unconstrained households:

$$\left| \frac{\partial k_{i2}}{\partial q_1} \frac{q_1}{k_{i2}} \right| < \left| \frac{\partial \overline{k}_2}{\partial q_1} \frac{q_1}{\overline{k}_2} \right|, \quad \forall i \in \mathcal{C}.$$

- Constrained households have few resources for the down payment
- At fixed q_1, more constrained HHs lower aggregate price sensitivity

◂ Proposition 2

Financial and Interest Rate Shocks

- A fall in θ directly reduces debt capacity and lowers the asset price
- Lower θ: more constrained HHs and lower consumption $\Rightarrow$ capital outflows
- Higher R: lower q_1 raises $\overline{w}$, while the direct interest-rate effect lowers it
- For R shocks, the constrained share and amplification have no general analytical sign
- Numerically, greater inequality amplifies consumption and price drops for both shocks

◂ Back ◂ TFP Shocks ▸ Financial Shock ▸ Interest Rate Shock

Model with Financial Shock

- Hold $z_1 = 1$ and let θ_1 fall from $\theta_H = 0.8$ to $\theta_L = 0.7$ in period 1

	Baseline	Alternative
Initial Wealth Gini	0.604	0.695
Period 1 Wealth Gini ($\theta_1 = \theta_H$)	0.570	0.644
Constrained Fraction at Period 1 ($\theta_1 = \theta_H$)	37.44%	49.36%
Constrained Fraction at Period 1 ($\theta_1 = \theta_L$)	45.52%	57.60%
Consumption change ($\theta_H \to \theta_L$)	-2.70%	-3.46%
Capital price change ($\theta_H \to \theta_L$)	-1.22%	-1.66%

◂ Back ◂ Main Slides

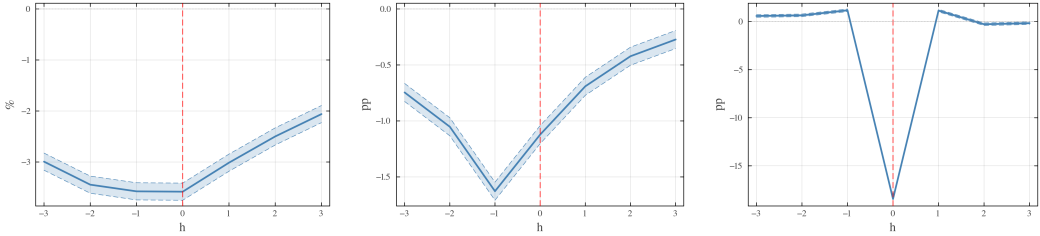
Model with Interest Rate Shock

- Hold $z_1 = 1$ and θ fixed, and let $R_1 \in \{R_h, R_l\}$ with $R_l = 1/\beta$ and a 2pp rise to R_h

	Baseline	Alternative
Initial Wealth Gini	0.604	0.695
Period 1 Wealth Gini ($R_1 = R_l$)	0.570	0.644
Constrained Fraction at Period 1 ($R_1 = R_l$)	37.71%	58.35%
Constrained Fraction at Period 1 ($R_1 = R_h$)	38.93%	61.63%
Consumption change ($R_l \to R_h$)	-2.35%	-2.43%
Capital price change ($R_l \to R_h$)	-1.98%	-2.00%

◂ Back ◂ Main Slides

Aggregate Shocks in Sudden Stops



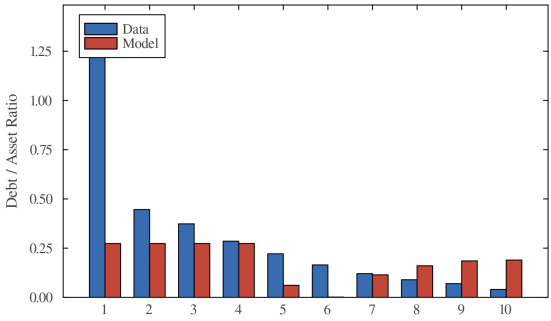
(a) Aggregate TFP (b) World Interest Rate (c) Borrowing Limit

Figure: Aggregate Shocks during Sudden Stops

Back

Leverage Ratio from Model

- Ex ante productivity generates U-shaped leverage ratio across distribution



Back

Sudden Stop Episodes

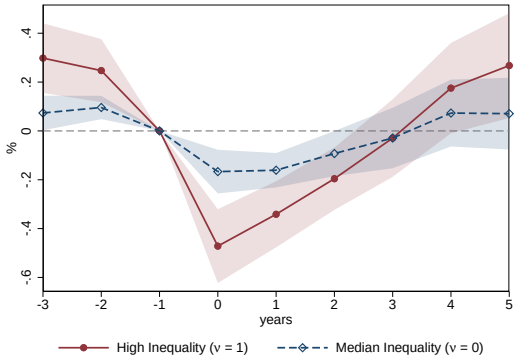
Country	Years	Country	Years	Country	Years
Algeria	1991, 1999	Germany	2009	Peru	1985
Argentina	1990, 2002	Greece	2012	Philippines	1984, 1998, 2009
Australia	2008	Hungary	1995, 2009	Poland	1999
Brazil	1984, 2003	Iceland	2001, 2009	Portugal	1982, 2012
Bulgaria	1994, 2009	Indonesia	1984, 1998	Russia	1999
Canada	1982	Italy	1983, 2012	South Africa	1983, 1985
Chile	1982	Japan	2015	Spain	2009, 2012
Colombia	1989, 1999	Korea	1998	Sweden	2001
Croatia	1998	Malaysia	1994	Switzerland	2010
Denmark	1999, 2010	Mexico	1983, 1995	Thailand	1998
Dominican Republic	1981, 1994	Morocco	1983, 2003	Türkiye	1994, 2001
Ecuador	1999	Netherlands	2003, 2009	Ukraine	1999
Egypt	1991	Nigeria	1992, 1995, 2000	United Kingdom	1991
El Salvador	1999	Norway	1990, 1999, 2008	Uruguay	1982, 2009
Finland	2000	Pakistan	2002, 2009	Venezuela	1983
France	1997	Panama	1995, 1998		

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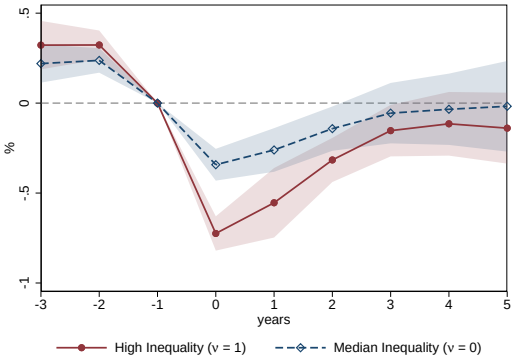
Dynamic Effects on GDP and Consumption

- Local projections interact sudden-stop outflows with lagged wealth inequality

$$\Delta y_{i,t+h} = \alpha_h + \beta_h \text{NetFlow}_{it}^{SS} + \gamma_h \text{NetFlow}_{it}^{SS} \nu_{i,t-1} + \Gamma_h X_{it} + FE + \varepsilon_{i,t+h}$$



(a) GDP per capita



(b) Consumption per capita

Back

Interactions with Other Country Characteristics

- Include interaction terms with country characteristics in the baseline regression

$$y_{it} = \beta \text{NetFlow}_{it}^{SS} + \gamma \text{NetFlow}_{it}^{SS} \times \nu_{i,t-1} + \sum_{x \in \text{Characteristics}} \left[\lambda^x \text{NetFlow}_{it}^{SS} \times x_{i,t-1} + \delta^x x_{i,t-1} \right] + FE + \epsilon_{it} \quad (1)$$

$$y_{it} = \beta \text{NetFlow}_{it}^{SS} + \gamma \text{NetFlow}_{it}^{SS} \times \nu_{i,t-1} + \kappa \text{NetFlow}_{it}^{SS} \times \text{GDP}_{i,t-1} + \sum_{x \in \text{Characteristics}} \left[\lambda^x \text{NetFlow}_{it}^{SS} \times x_{i,t-1} + \delta^x x_{i,t-1} \right] + FE + \epsilon_{it} \quad (2)$$

Estimates of γ	GDP	Consumption	Investment	Stock prices
Regression (1)	-0.512^{***}	-0.329^{***}	-1.587^{*}	-5.063^{**}
	(0.093)	(0.057)	(0.818)	(1.851)
Regression (2)	-0.546^{***}	-0.340^{***}	-1.701^{***}	-5.674^{***}
	(0.114)	(0.070)	(0.477)	(2.004)
Observations	864	864	847	782

Back

Wealth Inequality and External Debt

- Relationship with $\log(\text{External Debt}/\text{External Assets})$ from Lane and Milesi-Ferretti, 2018

	Within (1)	Within (2)	Between (3)	Within (4)	Within (5)	Between (6)
Wealth Gini$_{t-1}$	1.503*	1.672**	2.431*	1.863***	1.722**	2.732**
	(0.820)	(0.764)	(1.257)	(0.585)	(0.678)	(1.251)
GDP$_{t-1}$				0.148	0.981*	-9.831
				(0.595)	(0.569)	(16.009)
Real exchange rate$_{t-1}$				-0.216	-0.135	0.836
				(0.265)	(0.221)	(1.179)
Trade openness$_{t-1}$				-0.002	0.003	-0.004
				(0.002)	(0.002)	(0.003)
Reserves/GDP$_{t-1}$				-1.138**	-0.907*	0.758
				(0.519)	(0.479)	(0.667)
Country FE	Yes	Yes	–	Yes	Yes	–
Year FE	No	Yes	–	No	Yes	–
R^2	0.012	0.018	0.056	0.096	0.073	0.119
Observations	1,093	1,093	52	1,093	1,093	52

Back

First-Order Conditions for the Two Taxes

- $\tilde{c}_{j,t} \equiv c_{j,t} - \nu(n_{j,t})$; $\lambda_{j,t}$: social marginal value of wealth
- $\gamma_{j,t}, \eta_{j,t}$: multipliers on investment and saving implementability

$$-\operatorname{Cov}_j(\lambda_t, k_t) + \int \gamma_{j,t-1} u'(\tilde{c}_{j,t})\, dj = 0$$

- Wealth tax balances redistribution and returns to capital chosen in $t-1$

$$\frac{1}{R_t} \operatorname{Cov}_j(\lambda_t, b_{t+1}) + \theta_t q_t \int \gamma_{j,t} u'(\tilde{c}_{j,t})\, dj - \int \eta_{j,t} u'(\tilde{c}_{j,t})\, dj = 0$$

- Debt tax balances redistribution and current investment and saving incentives

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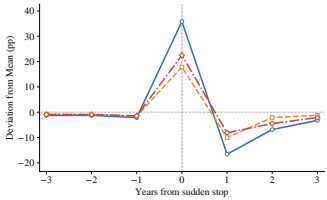
Parameter Calibration

Parameter	Value	Source
σ	2	Standard
β	0.91	Standard
δ	0.08	Standard
ψ	2	Standard
α	0.34	Labor income share of 66%
ϕ	2.42	Penn World Tables 11.0
ρ_z	0.84	Penn World Tables 11.0
σ_z	0.029	Penn World Tables 11.0
R	1.011	Bianchi and Mendoza, 2018
ρ_R	0.68	Bianchi and Mendoza, 2018
σ_R	0.0186	Bianchi and Mendoza, 2018
ζ	0.8363	Estimated
θ_H	0.291	Estimated
θ_L	0.095	Estimated
p_{HH}	0.899	Estimated
σ_ω	0.8636	Estimated

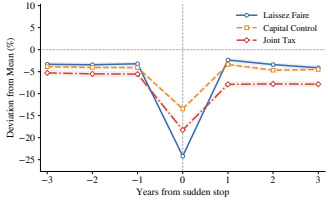
▸ Back

Dynamics under Taxation

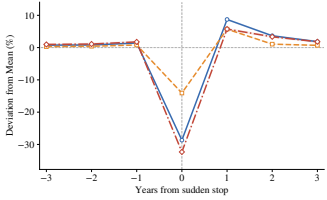
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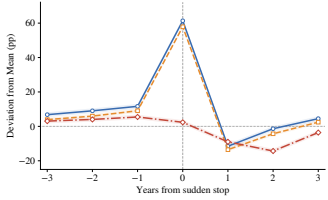
(a) Current Account / GDP



(b) Consumption



(c) Asset Price



(d) Constrained Fraction